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RESEARCH-ARTICLE

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Blockchain Technology Enabling Applications in Smart Agriculture

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Abstract

Blockchain technology is a decentralized, highly secure and data tamper-proof information storage and transmission technology, which ensures the integrity and trustworthiness of data through cryptographic principles. With the vigorous development of science and technology, blockchain technology is gradually penetrating and deeply integrating into various industries. The digitization process in rural areas is relatively lagging behind, and the roots of smart agriculture in rural areas are still shallow, resulting in the development of blockchain technology in the field of smart agriculture, there are problems such as complex data sources, irregular data transmission, and shortage of composite talents in blockchain technology and agriculture, which not only restricts the development of blockchain technology, but also affects the efficiency of agricultural production. Therefore, based on the perspective of blockchain technology and smart agriculture, this paper describes the practical application of blockchain technology in smart agriculture, which has certain practical significance for promoting the high-quality development of agriculture and the modernization of agriculture and rural areas.

CCS Concepts

• **Social and professional topics** → Professional topics; Management of computing and information systems.

Keywords

blockchain technology, smart agriculture, chain management

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1 Introduction

As a cutting-edge concept and innovative mode of modern agricultural development, smart agriculture makes use of modern information technology, Internet of Things, big data and artificial intelligence to realize intelligent, precise and efficient management of the whole process of agricultural production [1]. This model not only significantly improves the efficiency of agricultural production, effectively reduces resource consumption, but also further optimizes the quality of agricultural products and the agricultural ecological environment, laying a foundation for the realization of sustainable agricultural development. Under the support of the Digital China development strategy and a series of policies, China's modern agriculture is moving towards smart agriculture characterized by information as a production factor, the application of the Internet, Internet of Things, big data, cloud computing, blockchain, artificial intelligence and smart equipment, and at the same time, smart agriculture is gradually leading the agricultural industry in the direction of higher quality, greener and smarter. Although smart agriculture has a good development prospect, at the same time, its development is also facing a series of problems such as lack of technological maturity, difficulty in data acquisition and integration, imperfect standard system, limited policy support, low market acceptance, shortage of professionals, and limitations in rural infrastructure and Internet coverage, which prevent a significant breakthrough in agricultural operation efficiency.

Blockchain technology is a new type of computer technology application mode based on distributed data storage, peer-to-peer transmission, consensus mechanism and encryption algorithm, which builds a shared database that is unforgeable, traceable, traceable, open, transparent and collectively maintained, and realizes distributed storage and verification of data through decentralization, thus improving data security and credibility, and shows great potential for application in many fields such as finance, Internet of Things, logistics, public management and so on. logistics, public management and other fields show great potential for application [2].

In the face of the urgent demand for sustainable development of global agriculture, smart agriculture, as a cutting-edge concept of modernized agriculture, urgently needs an efficient, safe and trustworthy technological means to support the intelligent, precise and efficient management of the whole process. Blockchain technology, with its decentralized, untamperable, full traceability and open and transparent characteristics, provides a powerful data security guarantee and trust building mechanism for smart agriculture. In smart agriculture, blockchain technology can integrate data

from all aspects of agricultural production, circulation and sales, and realize the openness, transparency and non-tamperability of information, so as to improve the quality and market competitiveness of agricultural products. At the same time, blockchain can also promote the development of agricultural intelligence and automation, reduce intermediate links, improve production efficiency and reduce costs [3]. The combination of smart agriculture and blockchain technology is an inevitable choice to comply with the global agricultural development trend and promote the agricultural industry in the direction of higher quality, greener and smarter.

2 Development status of smart agriculture

2.1 The concept of smart agriculture

Smart Agriculture refers to the combination of modern science and technology and agricultural cultivation, through the application of modern information technology, Internet of Things technology, artificial intelligence and other high-tech means, the traditional agricultural production of digital, networked, intelligent transformation, so as to achieve unmanned, automated, intelligent management of the new agricultural form [4]. It is one of the important directions for the development of modern agriculture, and promotes the transformation, upgrading and sustainable development of agricultural production through scientific and technological innovation and the application of information technology. Compared with traditional agriculture, smart agriculture has the following characteristics:

(1) Optimization of production links: in the process of planting or farming, the smart agriculture system can monitor the environmental parameters of the farmland or farm in real time, which not only realizes the automation and refinement of the production process, but also significantly improves the efficiency of resource utilization and reduces production costs. At the same time, the smart agriculture platform can also scientifically manage the crop growth cycle, optimize the planting or breeding plan, and ensure the quality and yield of agricultural products.

(2) Automated production: Smart agriculture realizes intelligent monitoring and automated operation of the whole process of agricultural production through the integration of advanced sensors, automation equipment, robotics and the Internet of Things platform. From seeding, irrigation, fertilization to pest control, harvesting and other links, all without or with little human intervention, the system automatically complete the decision-making and implementation, greatly improving production efficiency, reducing labor costs, while ensuring the precise control of the growing environment of crops and the effective use of resources.

(3) Data management: Smart agriculture uses sensors installed in the farmland to collect real-time data such as soil moisture, temperature, nutrient content and crop growth conditions, and transmits them to cloud servers through wireless networks for storage and analysis. For example: 1) Internet of Things (IoT) technology. It is one of the core technologies of smart agriculture, through sensors, intelligent devices and cloud computing and other technical means, it can monitor environmental parameters such as soil moisture, temperature, air humidity, etc., and transmit these data to the cloud to help farmers in decision-making and resource management; 2) UAV technology. Used for crop pest monitoring and control, farmland mapping, precision fertilization and many

other aspects, through the high-definition camera and multispectral sensors carried by the drone, to obtain high-definition images of farmland and multispectral data, and then analyze the growth status of crops, pests and diseases, and other information.

2.2 Realistic dilemmas facing smart agriculture

The development of smart agriculture cannot be separated from the comprehensive support of data, covering soil monitoring, meteorological prediction, environmental investigation, market adjustment, logistics and transportation, and other dimensions, the taking, processing, analysis, sharing and efficient use of these data constitutes the core link of the smart agriculture system. While China's rural informatization construction is in the development stage, the sharing platform of data is not yet sound. Some agricultural enterprises, in the pursuit of technological innovation and business expansion, may have neglected to comprehensively assess and prevent the risk of information security across the industrial chain. Agricultural production subjects often have insufficient understanding of the importance of information security and lack the necessary protective awareness and measures [5].

2.2.1 Data Collection. In the data collection link, smart agriculture faces the real problems of limited data sources, uneven quality and high collection costs. The relatively weak infrastructure in rural areas and the lack of sufficient sensors and IoT devices lead to a limited scope of data collection. At the same time, due to differences in equipment quality, maintenance status and environmental factors, the quality of collected data is difficult to guarantee, affecting the accuracy of subsequent analysis. High-quality data collection equipment requires professional technicians to install, debug and maintain, and for small-scale farmers, such high input costs are often difficult to bear, resulting in their inability to enjoy the convenience and benefits of smart agriculture, which not only restricts the popularization of data collection, but also hinders the further promotion and application of smart agriculture technology.

2.2.2 Data processing and analyzing link. In this link, smart agriculture is faced with the reality of complex data processing, high requirements for analysis technology specialization, and the difficulty of transforming the application of analysis results. The data source has a wide range of structural differences, and the accumulation of data over a long period of time requires the use of highly specialized algorithms and models for efficient cleaning, integration and in-depth mining, however, the current technology is still difficult to cope with these complex needs, resulting in a cumbersome data processing process and efficiency limitations. The specialized threshold of analytics technology makes it difficult for general practitioners to conduct high-quality data analysis independently. However, even if valuable analysis results are successfully obtained, how to effectively translate them into improvement measures in actual agricultural production is still an urgent problem to be solved.

2.2.3 Data Sharing. The relative lag in the construction of informationization in rural areas not only leads to uneven levels of technical protection of smart agriculture systems, but also farmers generally have a weak awareness of data privacy protection, making it difficult to effectively resist security threats such as hacking

and data theft. Although the country has introduced a series of data protection regulations, the relevant laws and regulations in the field of smart agriculture, especially for the protection of privacy in agricultural data, are not yet sufficiently complete and the enforcement of them needs to be strengthened. In addition, the development of smart agriculture requires data sharing, and the current data sharing mechanism is not yet perfect, and the privacy protection measures are insufficient, which may easily lead to the phenomenon of data being abused or leaked in the sharing process.

3 Current status of blockchain technology application in smart agriculture

In the process of smart agriculture production, it involves many aspects such as data adoption, processing, analysis, sharing and efficient utilization, etc. The limitations of the traditional agricultural data management model, such as the single-point vulnerability brought by the centralized architecture, the vulnerability of data to tampering and the potential risk of information leakage, are gradually restricting the trustworthiness and value of the utilization of agricultural data, and at the same time, it also puts forward a higher demand for the security of the data and traceability ability. In the face of these challenges, the introduction of blockchain technology brings solutions for smart agriculture. Blockchain provides a more reliable infrastructure for agricultural data storage, transmission and management with decentralized architecture, distributed ledger mechanism and high data security [6,7,8]. It not only effectively disperses the system risk and reduces the impact of single-point failure on the overall data integrity, but also guarantees the tamperability and transparency of data through encryption algorithms and consensus mechanisms, which provides a solid technical support for the full traceability of agricultural products. The integration mode of “blockchain + agriculture” is gradually becoming a key force to promote the high-quality development of smart agriculture, and smart agriculture will usher in a broader development prospect [9].

3.1 Blockchain and Traceability of Agricultural Products.

Blockchain technology builds a highly transparent, secure and tamper-proof traceability system in the traceability of agricultural products, which can be reflected in the blockchain from the planting of crops to the transportation and sale of agricultural products, ensuring the accuracy and completeness of the data. At the initial stage of crop cultivation, blockchain is closely integrated with IoT technology, and key data such as soil humidity, light intensity, and fertilizer application are monitored and recorded in real time through smart sensors. After encrypted processing, these data are securely stored on the blockchain to form a tamper-proof timestamp record, while smart contracts automatically verify the accuracy of these data to ensure the authenticity and transparency of the planting process; the blockchain continues to play its role as it enters the stage of harvesting, processing and packaging of agricultural products. The harvest date, processing process, raw materials used, packaging methods, etc., are all recorded in detail on the blockchain, and consumers can easily access the full life cycle information of the product by simply scanning the traceability code on the product; in

the storage and transportation segment, IoT devices continuously monitor the storage conditions of the agricultural products, such as temperature, humidity, etc., and upload these real-time data to the blockchain. The immutability of the blockchain ensures that any anomalies during storage and transportation can be accurately recorded, facilitating problem tracking and responsibility definition; ultimately, in the agricultural product sales phase, blockchain technology provides a transparent supply chain management system. From producers to distributors to retailers, the transaction information of each link is recorded on the blockchain, including transportation details, sales locations and times, etc. This not only helps consumers understand the origin and flow of products, but also provides strong data support for regulators to facilitate their quality tracing and supervision.

3.2 Blockchain and Agricultural Financial Services.

Blockchain technology in agricultural financial services builds a secure and transparent operation platform for key aspects such as agricultural loans, insurance claims, investment management and capital operation through its non-tampering and decentralized characteristics. In the field of agricultural loans, financial institutions are able to utilize agricultural data recorded on the blockchain, such as crop yields, market trends, and meteorological conditions, to conduct risk assessments and provide farmers with tailor-made loan solutions. The introduction of smart contracts realizes the automation of loan issuance, and once the preset conditions are met, the funds can be quickly put in place, which greatly shortens the cycle of farmers' access to funds; in the area of insurance claims, by connecting the insurance contract with the data of IoT devices, such as meteorological stations, the smart contract is able to real-time monitor and automatically respond to specific climatic events such as droughts, floods, etc., and immediately start the claims process once the claims conditions are triggered. In terms of investment management and capital operation, all transaction records are encrypted and stored on the distributed ledger, and any participant can access the relevant data through authorization, ensuring the transparency and traceability of capital flows and promoting the effective allocation of agricultural capital.

3.3 Blockchain and Agricultural Insurance.

Blockchain technology can record the whole process of agricultural insurance transactions, including insured information, compensation records, and the status of the subject matter, etc., which are open, transparent and tamper-proof, enabling insurance companies, farmers and regulators to easily access and verify the data, which enhances the degree of trust; blockchain can provide real-time and accurate data (e.g., weather, market prices, crop conditions, etc.), helping insurance companies more accurately assess risks and formulate more targeted insurance products and rates; it realizes instant arrival of cross-border payments, reduces intermediary fees in cross-border payments, improves the processing efficiency of agricultural insurance cross-border business, and reduces transaction costs.

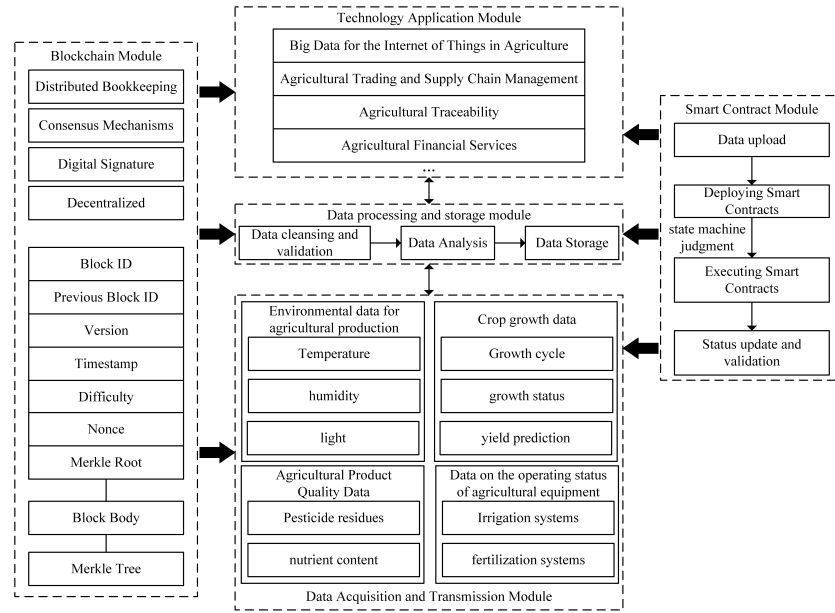


Figure 1: Application framework of blockchain technology in the field of smart agriculture.

3.4 Blockchain and Agricultural Supply Chain.

Blockchain technology allows information from every link in the chain, from the cultivation, processing, transportation to the sale of agricultural products, to be securely and irrevocably recorded on the chain [10]. This means that consumers can obtain complete information about the product from source to terminal simply by scanning the traceability code or QR code on the product package, which only enhances consumers' trust in the product and facilitates efficient quality monitoring and safety tracing by the regulatory authorities; in the traditional agricultural supply chain, there is often a lack of trust between the participating entities due to information asymmetry, and the decentralized nature of blockchain creates a trust mechanism between the participants to establish a trust mechanism, and farmers, processors, logistics and transportation companies, retailers, etc. can realize closer cooperation based on data sharing on the blockchain; support the writing and execution of smart contracts, which can automatically execute certain business processes in the supply chain, and can ensure that every link in the supply chain is executed in accordance with the established rules and conditions; blockchain technology combined with cloud-based robots and other advanced technology, it can carry out real-time monitoring of the production, harvesting, transportation and other processes of agricultural products. For example, through the blockchain technology query, it can accurately understand the actual harvested area, number of harvests, and transportation trajectory of the agricultural products, thus effectively preventing the occurrence of fraudulent behaviors.

4 Blockchain Technology Enabling the Convergence of Smart Agriculture

4.1 Application Framework

The application framework of blockchain technology in smart agriculture is a multifaceted and cross-disciplinary systematic project that aims to utilize the characteristics of blockchain technology to enhance the efficiency and transparency of agricultural production, management and supply chain. As shown in Figure 1, the framework not only covers the collection, integration and storage of data, but also involves data processing, the execution of smart contracts, and the application of the technology in various aspects of agriculture. By integrating multiple modules and blockchain technology, it brings innovative management and operation modes to the agricultural field and promotes the modernization and intelligence of agriculture.

4.2 Data Acquisition and Transmission Module

In the system of blockchain technology-enabled smart agriculture, the data collection and transmission module is a crucial and fundamental link, which is responsible for real-time and accurate collection of various types of data in the process of agricultural production, and transmits these data to the blockchain module for further processing. Data collection methods mainly include: (1) Real-time collection of physical parameters of the agricultural production environment through various types of IoT sensors (such as temperature, humidity, light, soil pH, water level, pesticide residue, and other sensors) deployed in agricultural fields, greenhouses, farms, and other agricultural production sites. (2) The use of RFID (radio frequency identification) technology for agricultural products or agricultural equipment affixed with electronic tags, to achieve its identity, location, status and other information automatic diagnosis

and collection. (3) Utilizing unmanned aerial vehicle and satellite remote sensing technology, it conducts regular inspections on large-area farmland and obtains data from high-altitude viewpoints such as crop growth conditions, disease and pest situations, and so on. Environmental data, crop growth data, agricultural product quality data, and agricultural equipment operation status data in agricultural production are collected, and the data collected by IoT sensors are mainly transmitted to data centers or cloud servers through wireless methods (such as Wi-Fi, LoRa, NB-IoT, etc.), and for some data transmission scenarios that require high stability and security, data transmission is carried out by wired methods (such as optical fiber Ethernet, etc.) for data transmission. During data transmission, encryption technology (TLS/SSL) is used to encrypt data to ensure the security and integrity of agricultural production data during transmission.

4.3 Data processing and storage module

The data processing and storage module is an important step in smart agriculture, which is responsible for receiving data from the data collection and transmission module, and carrying out cleaning, checking, analyzing and aggregation processing to ensure the integrity and accuracy of the data. Using the distributed ledger feature of blockchain, the data is stored in multiple nodes in a decentralized manner, and each node keeps a complete copy of the data. Through the hash algorithm and timestamp mechanism, secure and tamper-proof storage of data is realized. Meanwhile, the privacy and access control of the data are guaranteed, and privacy protection technologies such as differential privacy and homomorphic encryption are used to encrypt the data or add noise to protect the personal privacy information from being leaked. The processed data not only supports accurate management and decision-making in agricultural production, but also promotes cross-organizational sharing of agricultural data, accelerating the digital transformation process of the agricultural industry.

4.4 Blockchain Module

The blockchain module is the core technology module in smart agriculture, which is a distributed database technology that stores data in the form of a chained data structure, where each data block is associated with the previous data block, forming a growing data chain. Each data block contains a certain amount of transaction information or other data, which is encrypted and verified before being added to the blockchain. In the field of smart agriculture, blockchain technology can realize the traceability and transparency of logistics and information flow, and improve the efficiency and credibility of the supply chain.

4.5 Smart Contract Module

Smart contract module is based on blockchain technology to realize the automated execution of contract terms, which can be automatically executed when pre-set conditions are met without human intervention, thus realizing the transparency, security and efficiency of the contract. In smart agriculture, smart contract modules are widely used in many aspects of agricultural production, management, and transactions to optimize processes, reduce costs, and improve efficiency and trust. In the automated management of

agricultural production, the smart contract module can automatically trigger irrigation, fertilization and other operations according to the environmental parameters of agricultural production (e.g., soil humidity, light intensity, etc.), realizing precise agricultural management; by setting the crop growth cycle and expected yield, the smart contract can automatically monitor and adjust the agricultural production plan to ensure the healthy growth of the crop; in the transaction and supply chain management of agricultural products, the smart contract in agricultural products trading and supply chain management, smart contracts can ensure the automatic implementation of trading terms, including the verification and confirmation of key information such as price, quantity, quality, etc. Through the non-tamperable records on the blockchain, smart contracts can track the circulation path of agricultural products and realize the full traceability from production to consumption. In agricultural financial services, smart contracts can be used in the field of agricultural insurance to automatically monitor and verify agricultural disaster situations, trigger the insurance compensation process, and improve the efficiency and fairness of insurance services. In agricultural loans, smart contracts can automatically assess loan risks and set repayment plans and interest rate conditions based on crop growth cycles and expected returns.

4.6 Technology Application Module

In the integration and development of blockchain technology-enabled smart agriculture, the technology application module covers the specific application of blockchain technology in various aspects of agricultural production, management, and sales, such as agricultural Internet of Things and big data, agricultural supply chain management, the traceability of the whole chain of agricultural products from production to consumption, agricultural insurance, agricultural loans, and other financial services, with the aim of improving the overall efficiency of agriculture, guaranteeing the quality and safety of agricultural products, and promoting agricultural sustainable development. In response to the demand for real-time monitoring and data analysis of the big data agricultural production environment, the IoT devices (sensors, RFID, etc.) are combined with blockchain technology to realize real-time monitoring and data collection of the agricultural production environment; blockchain technology ensures the authenticity and tamperability of the sensor data, providing accurate and intelligent decision-making support for the agricultural production; and the blockchain-based big data of agriculture is subjected to data mining and analysis, revealing the laws of agricultural production, optimizing resource allocation and improving production efficiency.

5 Conclusion

Blockchain technology has shown great potential and value in the integration and development of smart agriculture, and this paper analyzes the application prospects of blockchain technology in the field of smart agriculture with regard to the dilemma of smart agriculture in terms of data security and traceability. The use of blockchain technology, with its centralized, non-tamperable, transparent and traceable characteristics, to provide reliable data support for agricultural production can not only effectively improve the precision and intelligence of agricultural production, but also enhance

the quality and safety traceability of agricultural products, optimize the management of the agricultural supply chain, and promote the innovation and development of agricultural financial services. Using the application of blockchain technology in the traceability of agricultural products, agricultural financial services, agricultural Internet of Things and big data, it can further promote the digital and intelligent transformation of the agricultural industry and inject new vitality into the sustainable development of agriculture.

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